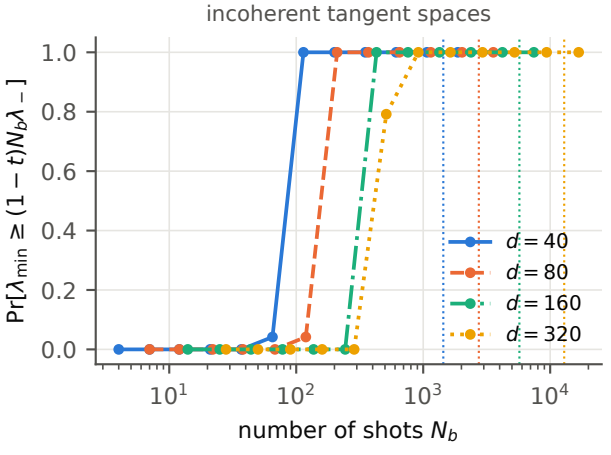
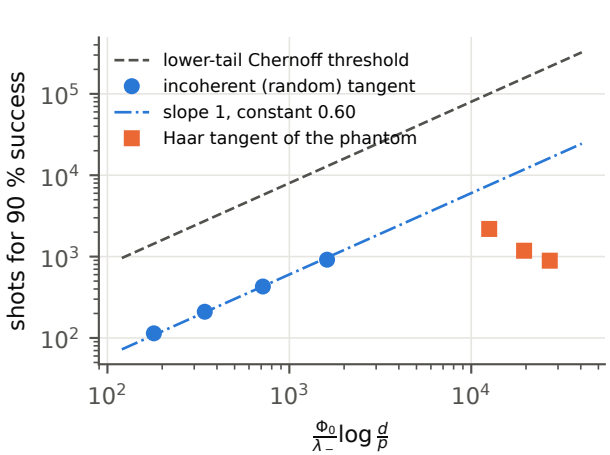

incoherent tangent spaces
1.0
0.8
0.6
0.4
0.2
0.0
$\Pr[\lambda_{\min} \geq (1-t)N_b\lambda_-]$
10^1
10^2
10^3
10^4
number of shots N_b
$d = 40$
$d = 80$
$d = 160$
$d = 320$


lower-tail Chernoff threshold
incoherent (random) tangent
slope 1, constant 0.60
Haar tangent of the phantom
shots for 90 % success
10^5
10^4
10^3
10^2
10^2
10^3
10^4
$\frac{\Phi_0}{\lambda_-}\log\frac{d}{p}$